

Lab 10 Report

AUTHOR

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```
library(bio3d)
library(ggplot2)
library(ggrepl)
```

Introduction to RCSB Protein Data Bank

PDB Statistics

```
pdb_stats <- read.csv("pdb_stats.csv")
pdb_stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	214078
2	1	0	13981
3	0	0	15759
4	3	1	4941
5	0	0	237
6	0	4	22

```
total_structures <- sum(pdb_stats$Total)

xray_total <- sum(pdb_stats$X.ray)
em_total <- sum(pdb_stats$EM)

percent_xray <- xray_total / total_structures * 100
percent_em <- em_total / total_structures * 100

percent_xray
```

```
[1] 80.95077
```

```
percent_em
```

```
[1] 12.83843
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

X-ray crystallography accounts for ~80.95% of PDB structures Electron microscopy accounts for ~12.84% of PDB structures

```
protein_rows <- grep("Protein", pdb_stats$Molecular.Type)
protein_total <- sum(pdb_stats$Total[protein_rows])

protein_proportion <- protein_total / total_structures
protein_percent <- protein_proportion * 100

protein_proportion
```

```
[1] 0.979118
```

```
protein_percent
```

```
[1] 97.9118
```

Q2: What proportion of structures in the PDB are protein?

The proportion of PDB structures that contain protein is approximately 0.979118, or 97.91%.

Q3: How many HIV-1 protease structures are in the current PDB?

After searching HIV-1 protease on the RCSB PDB website, I found that there are 1,227 HIV-1 protease structures.

Visualizing HIV-1 Protease Structure

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

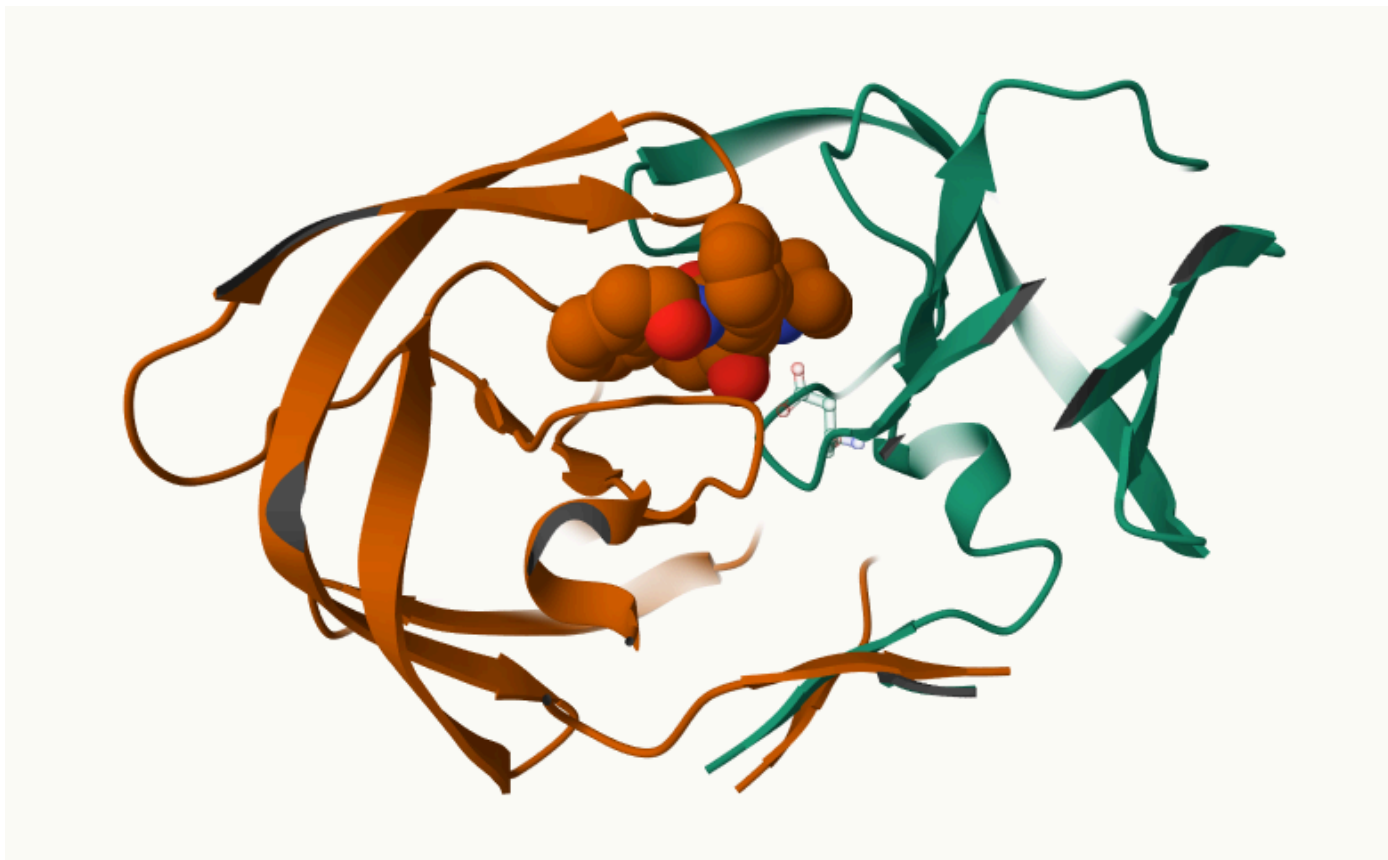
In X-ray crystal structures, hydrogen atoms are often not visible because they scatter X-rays weakly. For water molecules, the oxygen atom is usually the only atom clearly modeled, so each water molecule appears as one atom.

Q5: Conserved water molecule in the binding site

The conserved water molecule in the HIV protease binding site is HOH 306.

Q6: HIV Protease figure The image shows the two chains of HIV protease, the bound ligand, the catalytic ASP 25 residues. I was not able to add a component for the conserved water molecule even after asking for help in piazza. Indinavir or larger ligands could enter the binding site through movement of the flexible flap regions of HIV protease. These flaps can open to allow substrate or inhibitor entry and then close over the ligand in the active site.

```
knitr::include_graphics("images/protein_structure.png.png")
```



Introduction to bio3d

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

Call: `read.pdb(file = "1hsg")`

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this PDB object.

Q8: Name one of the two non-protein residues.

One non-protein residue is HOH, which represents water. Another is MK1, the ligand.

Q9: How many protein chains are in this structure?

There are 2 protein chains, chain A and chain B.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

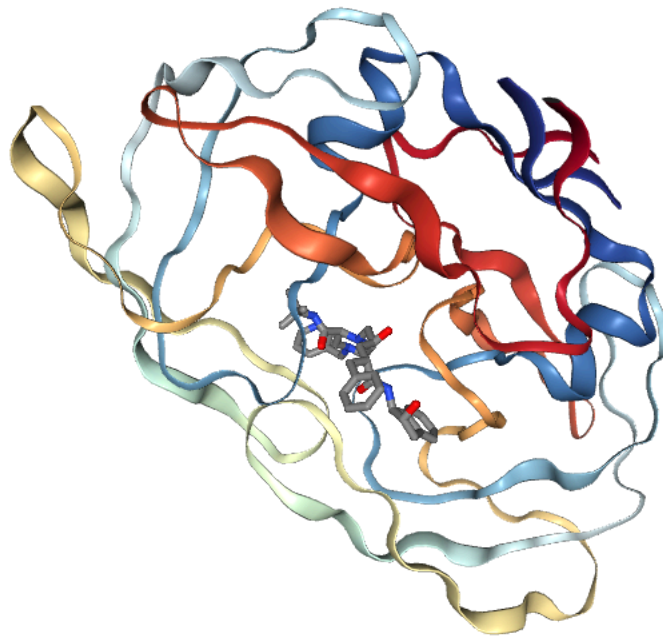
	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

PDB Visualization

```
# Optional interactive visualization
# install.packages("remotes")
# remotes::install_github("bioboot/bio3dview")
# install.packages("NGLViewerR")

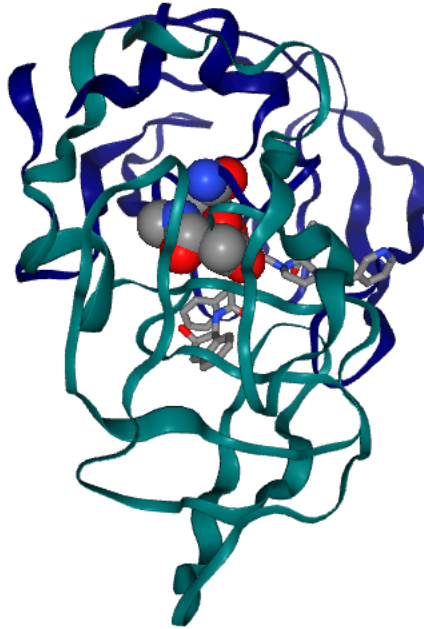
library(bio3dview)
library(NGLViewerR)

view.pdb(pdb) |>
  setSpin()
```



```
sele <- atom.select(pdb, resno = 25)

view.pdb(
  pdb,
  cols = c("navy", "teal"),
  highlight = sele,
  highlight.style = "spacefill"
) |>
  setRock()
```



Predicting Functional Motions of a Single Structure

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: `read.pdb(file = "6s36")`

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGMDLRAAVKSGSELGKQAKDIMDAGKLVT

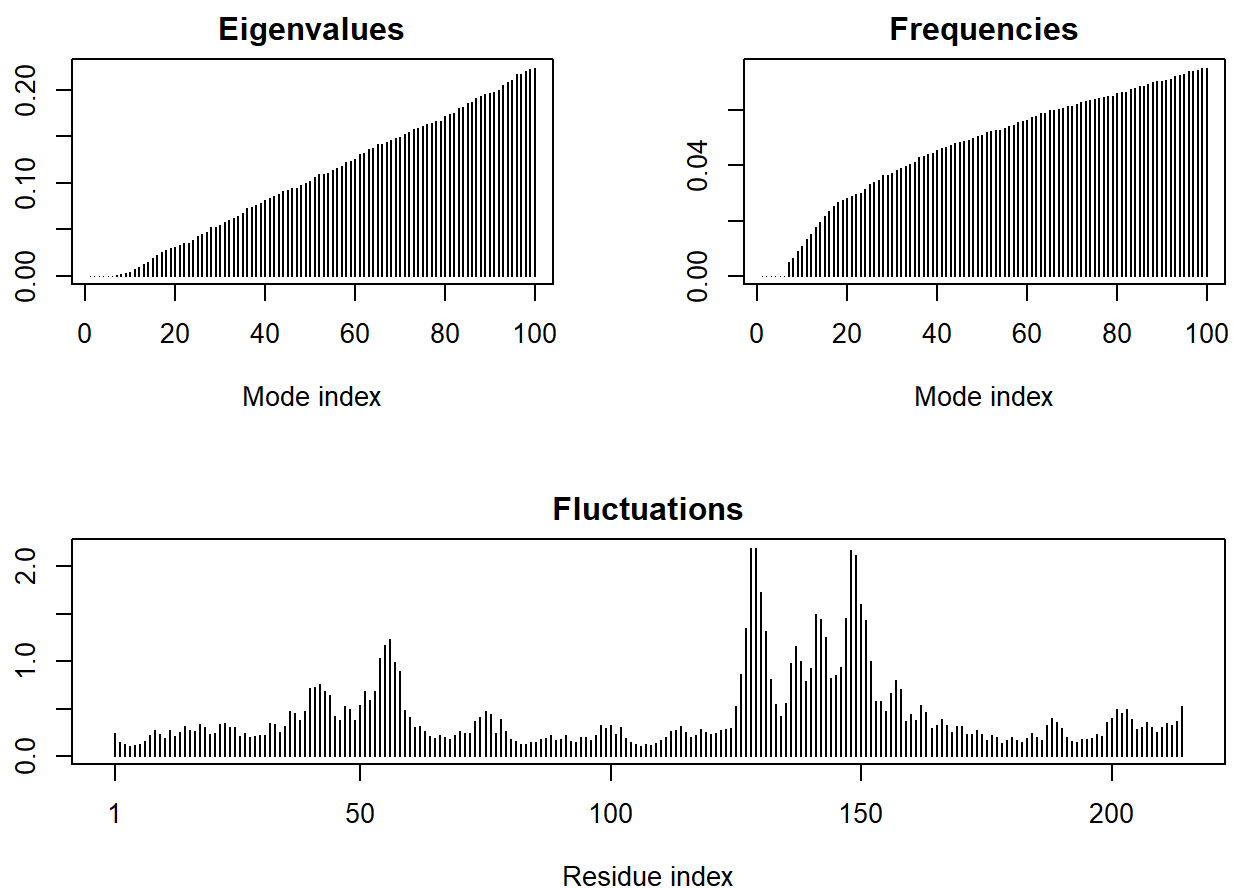
```
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.04 seconds.  
Diagonalizing Hessian... Done in 0.35 seconds.
```

```
plot(m)
```



```
mktrj(m, file = "adk_m7.pdb")
```

Comparative Structure Analysis of Adenylate Kinase

```
library(bio3d)  
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

      121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q10: Which package is found only on BioConductor and not CRAN?

The msa package is found on BioConductor and not CRAN.

Q11: Which package is not found on BioConductor or CRAN?

The bio3dview package is installed from GitHub, not CRAN or BioConductor.

Q12: True or False? Functions from the pak package can install packages from GitHub and BitBucket?

True.

Q13: How many amino acids are in this sequence?

The 1AKE chain A sequence has 214 amino acids.

Search and Retrieve ADK Structures

```
hits <- NULL
hits$pdb.id <- c(
  "1AKE_A", "6S36_A", "6RZE_A", "3HPR_A", "1E4V_A",
  "5EJE_A", "1E4Y_A", "3X2S_A", "6HAP_A", "6HAM_A",
  "4K46_A", "3GMT_A", "4PZL_A"
)

hits$pdb.id
```

```
[1] "1AKE_A" "6S36_A" "6RZE_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A"
[9] "6HAP_A" "6HAM_A" "4K46_A" "3GMT_A" "4PZL_A"
```

```
files <- get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE)
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb exists. Skipping download

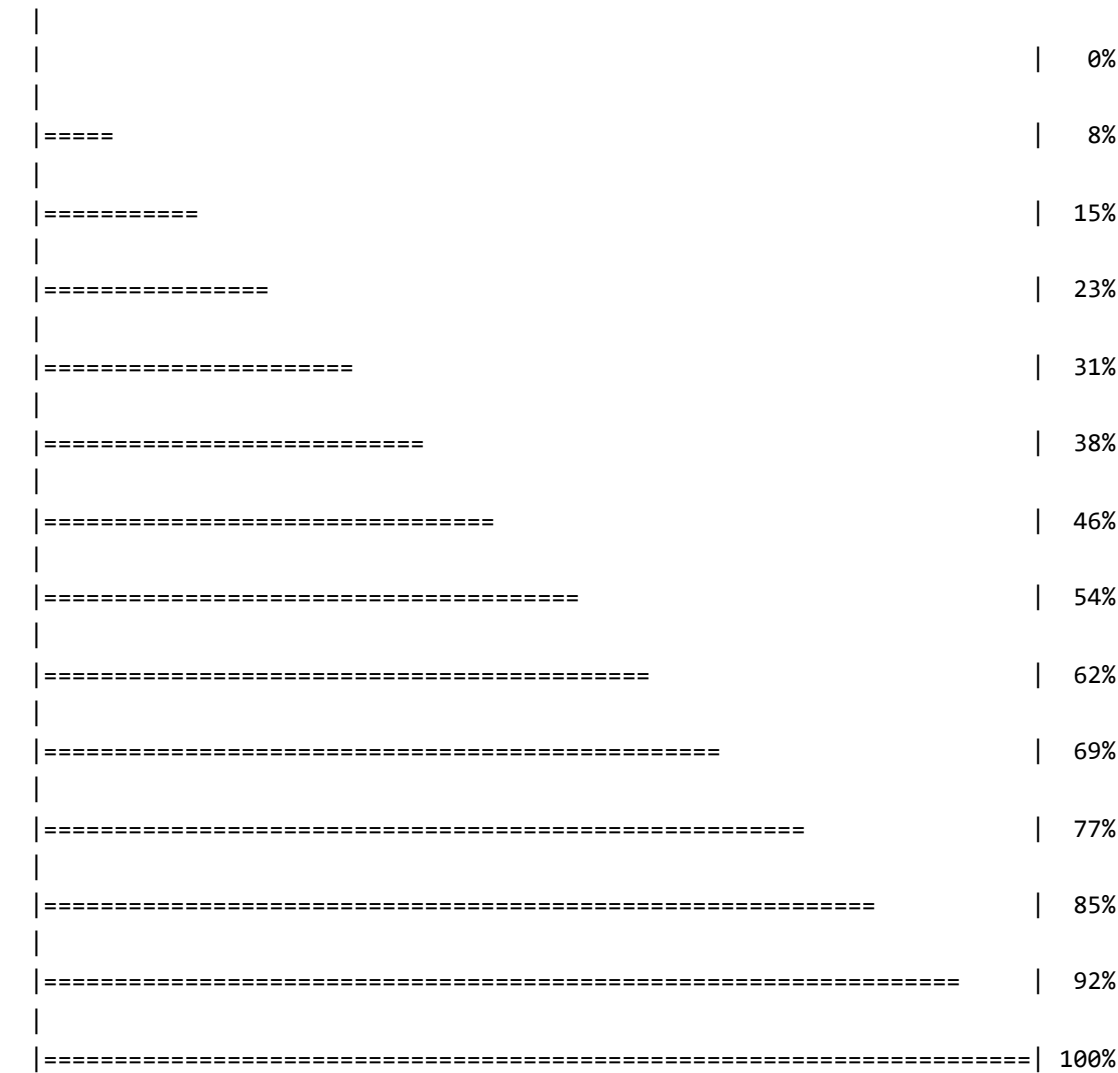
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb exists. Skipping download



files

- [1] "pdbs/split_chain/1AKE_A.pdb" "pdbs/split_chain/6S36_A.pdb"
- [3] "pdbs/split_chain/6RZE_A.pdb" "pdbs/split_chain/3HPR_A.pdb"
- [5] "pdbs/split_chain/1E4V_A.pdb" "pdbs/split_chain/5EJE_A.pdb"
- [7] "pdbs/split_chain/1E4Y_A.pdb" "pdbs/split_chain/3X2S_A.pdb"
- [9] "pdbs/split_chain/6HAP_A.pdb" "pdbs/split_chain/6HAM_A.pdb"
- [11] "pdbs/split_chain/4K46_A.pdb" "pdbs/split_chain/3GMT_A.pdb"
- [13] "pdbs/split_chain/4PZL_A.pdb"

Align and Superpose Structures

```
pdb<- pdbaln(files, fit = TRUE, exefile = "msa")
```

Reading PDB files:

```
pdb<-pdbs/split_chain/1AKE_A.pdb
pdb<-pdbs/split_chain/6S36_A.pdb
pdb<-pdbs/split_chain/6RZE_A.pdb
pdb<-pdbs/split_chain/3HPR_A.pdb
pdb<-pdbs/split_chain/1E4V_A.pdb
pdb<-pdbs/split_chain/5EJE_A.pdb
pdb<-pdbs/split_chain/1E4Y_A.pdb
pdb<-pdbs/split_chain/3X2S_A.pdb
pdb<-pdbs/split_chain/6HAP_A.pdb
pdb<-pdbs/split_chain/6HAM_A.pdb
pdb<-pdbs/split_chain/4K46_A.pdb
pdb<-pdbs/split_chain/3GMT_A.pdb
pdb<-pdbs/split_chain/4PZL_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
.. PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

```
seq1<-pdbs/seq:1 name: pdbs/split_chain/1AKE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq2<-pdbs/seq:2 name: pdbs/split_chain/6S36_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq3<-pdbs/seq:3 name: pdbs/split_chain/6RZE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq4<-pdbs/seq:4 name: pdbs/split_chain/3HPR_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq5<-pdbs/seq:5 name: pdbs/split_chain/1E4V_A.pdb
seq6<-pdbs/seq:6 name: pdbs/split_chain/5EJE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq7<-pdbs/seq:7 name: pdbs/split_chain/1E4Y_A.pdb
seq8<-pdbs/seq:8 name: pdbs/split_chain/3X2S_A.pdb
seq9<-pdbs/seq:9 name: pdbs/split_chain/6HAP_A.pdb
seq10<-pdbs/seq:10 name: pdbs/split_chain/6HAM_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq11<-pdbs/seq:11 name: pdbs/split_chain/4K46_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
seq12<-pdbs/seq:12 name: pdbs/split_chain/3GMT_A.pdb
seq13<-pdbs/seq:13 name: pdbs/split_chain/4PZL_A.pdb
```

```
pdb<-
```

```
1 . . . 40
[Truncated_Name:1]1AKE_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:2]6S36_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:3]6RZE_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:4]3HPR_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:5]1E4V_A.pdb -----MRIILLGAPVAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:6]5EJE_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:7]1E4Y_A.pdb -----MRIILLGALVAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:8]3X2S_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:9]6HAP_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:10]6HAM_A.pdb -----MRIILLGAPGAGKGTQAQFIMEKYGIPQIS
[Truncated_Name:11]4K46_A.pdb -----MRIILLGAPGAGKGTQAQFIMAKFGIPQIS
[Truncated_Name:12]3GMT_A.pdb -----MRLILLGAPGAGKGTQANFIKEKFGIPQIS
[Truncated_Name:13]4PZL_A.pdb TENLYFQSNAMRIILLGAPGAGKGTQAKIIEQKYNIAHIS
                        **^*****      *  ^ *  **
1 . . . 40

41 . . . 80
[Truncated_Name:1]1AKE_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:2]6S36_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:3]6RZE_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:4]3HPR_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:5]1E4V_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:6]5EJE_A.pdb TGDMLRAAVKSGSELGKQAKDIMDACKLVDELVIALVKE
[Truncated_Name:7]1E4Y_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVKE
[Truncated_Name:8]3X2S_A.pdb TGDMLRAAVKSGSELGKQAKDIMDCGKLVDELVIALVKE
[Truncated_Name:9]6HAP_A.pdb TGDMLRAAVKSGSELGKQAKDIMDAGKLVDELVIALVRE
[Truncated_Name:10]6HAM_A.pdb TGDMLRAAIKSGSELGKQAKDIMDAGKLVDEIIIALVKE
[Truncated_Name:11]4K46_A.pdb TGDMLRAAIKAGTELGQAKSVIDAGQLVSDDIILGLVKE
[Truncated_Name:12]3GMT_A.pdb TGDMLRAAVKAGTPLGVEAKTYMDEGKLPDSLIIIGLVKE
[Truncated_Name:13]4PZL_A.pdb TGDMIRETIKSGSALGQELKKVLDAGELVSDEFIIKIVKD
                        ****^*  ^*  ^*  *  ^*  **  *  ^^  ^*^^
41 . . . 80

81 . . . 120
[Truncated_Name:1]1AKE_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:2]6S36_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:3]6RZE_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:4]3HPR_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:5]1E4V_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:6]5EJE_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:7]1E4Y_A.pdb RIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:8]3X2S_A.pdb RIAQEDSRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:9]6HAP_A.pdb RICQEDSRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:10]6HAM_A.pdb RICQEDSRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFD
[Truncated_Name:11]4K46_A.pdb RIAQDDCAKGFLDGFPR TIPQADGLKEVGVVVDYVIEFD
[Truncated_Name:12]3GMT_A.pdb RLKEADCANGYLF DGFPR TIPQADAMKEAGVAIDYVLEID
[Truncated_Name:13]4PZL_A.pdb RISKNDCCNGFLLDGVPR TIPQAQELDKLGVNIDYIVEVD
                        *^  *  ^*  **  ****  *  ^  *^  ^*^^*  *
```

```
121 . . . 160
[Truncated_Name:1]1AKE_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:2]6S36_A.pdb VPDELIVDKIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:3]6RZE_A.pdb VPDELIVDAIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:4]3HPR_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDGTG
[Truncated_Name:5]1E4V_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:6]5EJE_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:7]1E4Y_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:8]3X2S_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:9]6HAP_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:10]6HAM_A.pdb VPDELIVDRIVGRRVHAPSGRVYHVKNPPKVEGKDDVTG
[Truncated_Name:11]4K46_A.pdb VADSVIVERMAGRRAHLASGRTYHNVNPPKVEGKDDVTG
[Truncated_Name:12]3GMT_A.pdb VPFSEIIERMSGRRTHPASGRTYHVKNPPKVEGKDDVTG
[Truncated_Name:13]4PZL_A.pdb VADNLLIERITGRIHPASGRTYHTKFNPPKVADKDDVTG
* ^^^ ^ *** * *** ** ^***** *** **
121 . . . 160

161 . . . 200
[Truncated_Name:1]1AKE_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:2]6S36_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:3]6RZE_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:4]3HPR_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:5]1E4V_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:6]5EJE_A.pdb EELTTRKDDQEECVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:7]1E4Y_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:8]3X2S_A.pdb EELTTRKDDQEETVRKRLCEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:9]6HAP_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:10]6HAM_A.pdb EELTTRKDDQEETVRKRLVEYHQM TAPLIGYYSKEAEAGN
[Truncated_Name:11]4K46_A.pdb EDLVIREDDKEETVLARLGVYHNQ TAPLIAYYGKEAEAGN
[Truncated_Name:12]3GMT_A.pdb EPLVQRDDDKKEETVKKRLDVYEAQTKPLITYYGDWARRGA
[Truncated_Name:13]4PZL_A.pdb EPLITRTDDNEDTVKQRLSVYHAQTAKLIDFYRNFSSNT
* * * ** ^ * ** * * ** ^*
161 . . . 200

201 . . 227
[Truncated_Name:1]1AKE_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:2]6S36_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:3]6RZE_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:4]3HPR_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:5]1E4V_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:6]5EJE_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:7]1E4Y_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:8]3X2S_A.pdb T--KYAKVDGTKPVAEVRADLEKILG-
[Truncated_Name:9]6HAP_A.pdb T--KYAKVDGTKPVCEVRADLEKILG-
[Truncated_Name:10]6HAM_A.pdb T--KYAKVDGTKPVCEVRADLEKILG-
[Truncated_Name:11]4K46_A.pdb T--QYLKFDGTKA VAESAELEKALA-
[Truncated_Name:12]3GMT_A.pdb E-----NGLKAPA-----YRKISG-
[Truncated_Name:13]4PZL_A.pdb KIPKYIKINGDQAVEKVSQDIFDQLNK
*
201 . . 227
```

Call:

```
pdbsaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

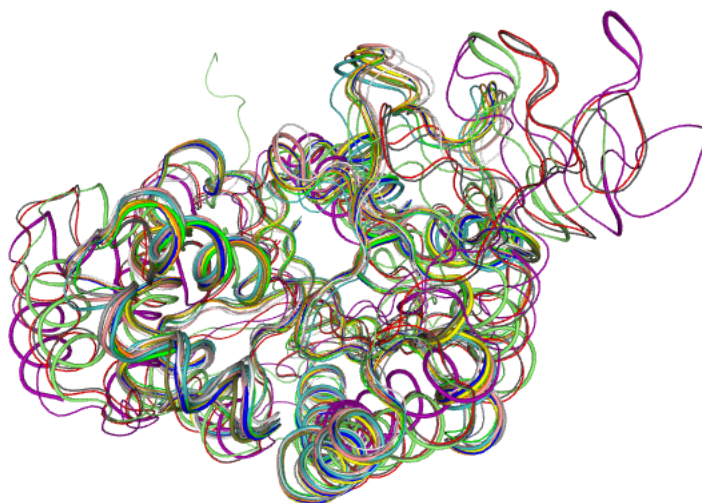
```
pdbs, fasta
```

Alignment dimensions:

```
13 sequence rows; 227 position columns (204 non-gap, 23 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
library(bio3dview)  
view.pdbs(pdbs)
```



Annotate Collected PDB Structures

```
ids <- basename.pdb(pdbs$id)  
anno <- pdb.annotate(ids)  
  
unique(anno$source)
```

- [1] "Escherichia coli"
- [2] "Escherichia coli K-12"
- [3] "Escherichia coli 0139:H28 str. E24377A"
- [4] "Escherichia coli str. K-12 substr. MDS42"
- [5] "Photobacterium profundum"
- [6] "Burkholderia pseudomallei 1710b"
- [7] "Francisella tularensis subsp. tularensis SCHU S4"

anno

structureId	chainId	macromoleculeType	chainLength	experimentalTechnique	
1AKE_A	1AKE	A	Protein	214	X-ray
6S36_A	6S36	A	Protein	214	X-ray
6RZE_A	6RZE	A	Protein	214	X-ray
3HPR_A	3HPR	A	Protein	214	X-ray
1E4V_A	1E4V	A	Protein	214	X-ray
5EJE_A	5EJE	A	Protein	214	X-ray
1E4Y_A	1E4Y	A	Protein	214	X-ray
3X2S_A	3X2S	A	Protein	214	X-ray
6HAP_A	6HAP	A	Protein	214	X-ray
6HAM_A	6HAM	A	Protein	214	X-ray
4K46_A	4K46	A	Protein	214	X-ray
3GMT_A	3GMT	A	Protein	230	X-ray
4PZL_A	4PZL	A	Protein	242	X-ray
resolution		scopDomain		pfam	
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
3X2S_A	2.80	<NA>	<NA>		
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)		
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3GMT_A	2.10	<NA>	<NA>		
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)		
ligandId					
1AKE_A	AP5				
6S36_A	CL (3),NA,MG (2)				
6RZE_A	NA (3),CL (2)				
3HPR_A	AP5				
1E4V_A	AP5				
5EJE_A	AP5,C0				
1E4Y_A	AP5				
3X2S_A	JPY (2),AP5,MG				
6HAP_A	AP5				
6HAM_A	AP5				
4K46_A	ADP,AMP,P04				

3GMT_A S04 (2)
4PZL_A CA,FMT,GOL

ligandName

1AKE_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6S36_A CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A SODIUM ION (3),CHLORIDE ION (2)
3HPR_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
1E4V_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
5EJE_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6HAM_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE
4K46_A ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A SULFATE ION (2)
4PZL_A CALCIUM ION,FORMIC ACID,GLYCEROL

source

1AKE_A Escherichia coli
6S36_A Escherichia coli
6RZE_A Escherichia coli
3HPR_A Escherichia coli K-12
1E4V_A Escherichia coli
5EJE_A Escherichia coli 0139:H28 str. E24377A
1E4Y_A Escherichia coli
3X2S_A Escherichia coli str. K-12 substr. MDS42
6HAP_A Escherichia coli 0139:H28 str. E24377A
6HAM_A Escherichia coli K-12
4K46_A Photobacterium profundum
3GMT_A Burkholderia pseudomallei 1710b
4PZL_A Francisella tularensis subsp. tularensis SCHU S4

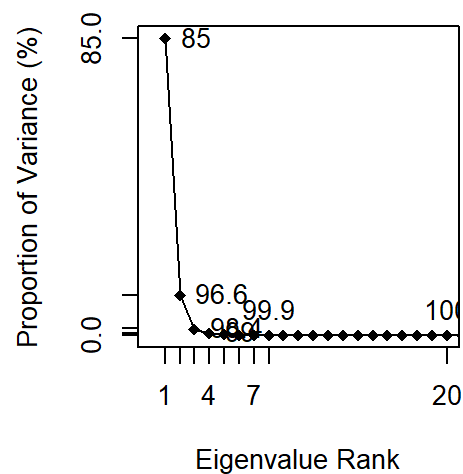
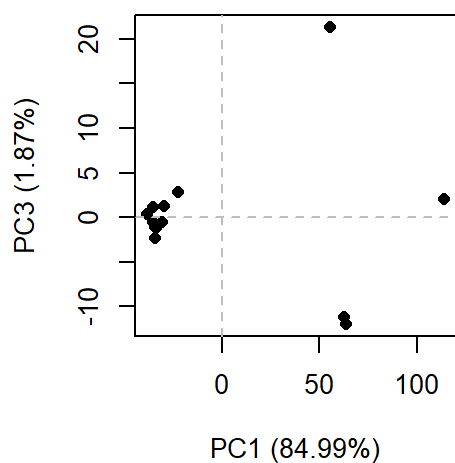
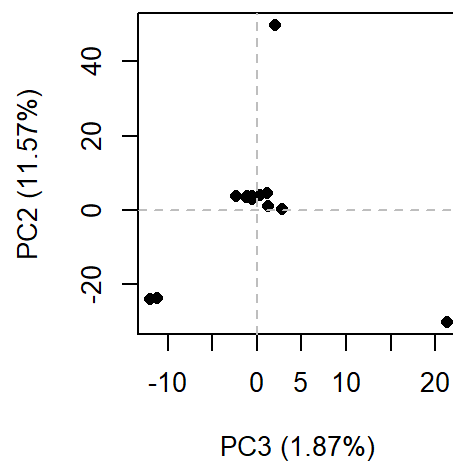
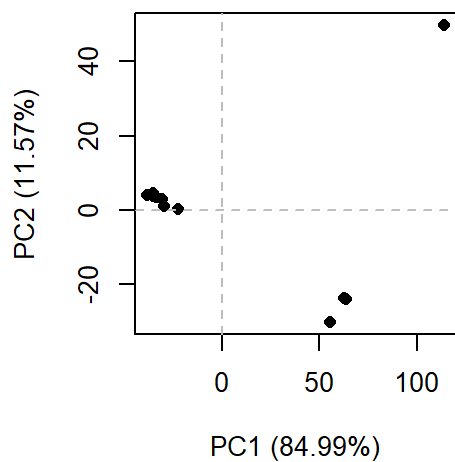
structureTitle

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBITOR AP5A REFINED AT 1.9 ANGSTROMS RESOLUTION: A MODEL FOR A CATALYTIC TRANSITION STATE
6S36_A Crystal structure of E. coli Adenylate kinase R119K mutant
6RZE_A Crystal structure of E. coli Adenylate kinase R119A mutant
3HPR_A Crystal structure of V148G adenylate kinase from E. coli, in complex with Ap5A
1E4V_A Mutant G10V of adenylate kinase from E. coli, modified in the Gly-loop
5EJE_A Crystal structure of E. coli Adenylate kinase G56C/T163C double mutant in complex with Ap5a
1E4Y_A Mutant P9L of adenylate kinase from E. coli, modified in the Gly-loop
3X2S_A Crystal structure of pyrene-conjugated adenylate kinase
6HAP_A Adenylate kinase
6HAM_A

Adenylate kinase			
4K46_A			
Crystal Structure of Adenylate Kinase from Photobacterium profundum			
3GMT_A			
Crystal structure of adenylate kinase from burkholderia pseudomallei			
4PZL_A			
structure of adenylate kinase from Francisella tularensis subsp. tularensis SCHU S4			
		citation	rObserved rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680
rWork spaceGroup			
1AKE_A	0.19600	P 21 2 21	
6S36_A	0.15940	C 1 2 1	
6RZE_A	0.18190	C 1 2 1	
3HPR_A	0.20620	P 21 21 2	
1E4V_A	0.19600	P 21 2 21	
5EJE_A	0.18630	P 21 2 21	
1E4Y_A	0.17800	P 1 21 1	
3X2S_A	0.20700	P 21 21 21	
6HAP_A	0.22370	I 2 2 2	
6HAM_A	0.20311	P 43	
4K46_A	0.16730	P 21 21 21	
3GMT_A	0.23500	P 1 21 1	
4PZL_A	0.19130	P 32	

Principal Component Analysis

```
pc.xray <- pca(pdbx)
plot(pc.xray)
```



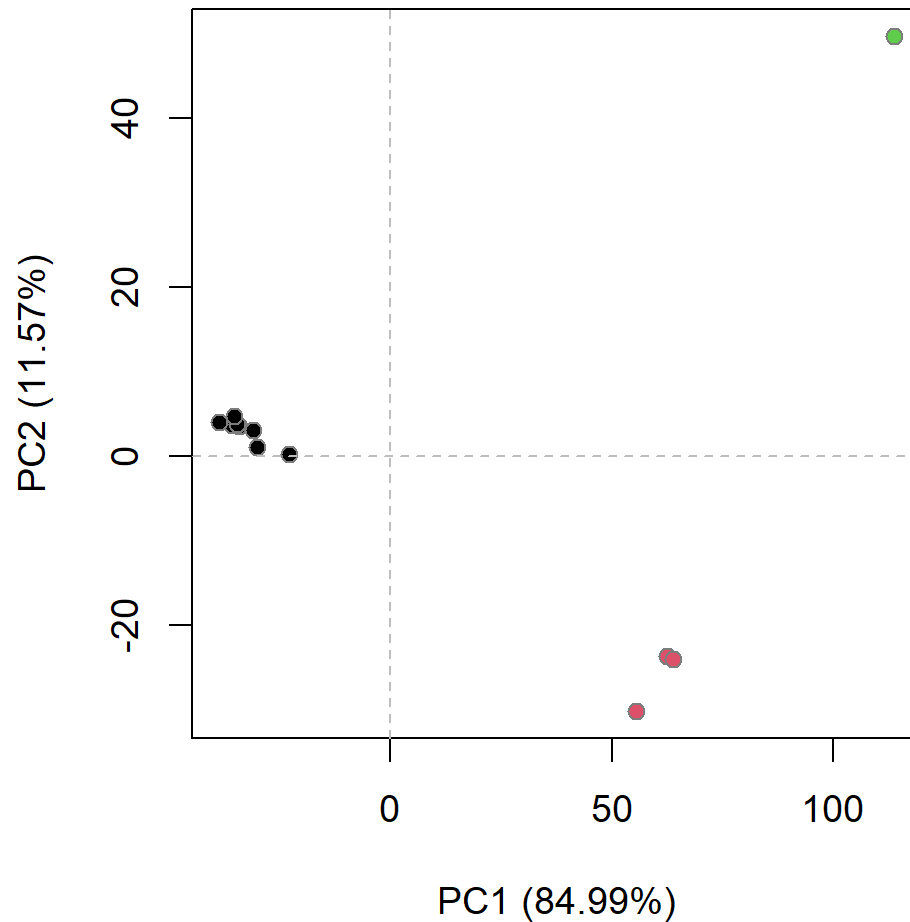
```
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

```
hc.rd <- hclust(dist(rd))

grps.rd <- cutree(hc.rd, k = 3)

plot(
  pc.xray,
  1:2,
  col = "grey50",
  bg = grps.rd,
  pch = 21,
  cex = 1
)
```

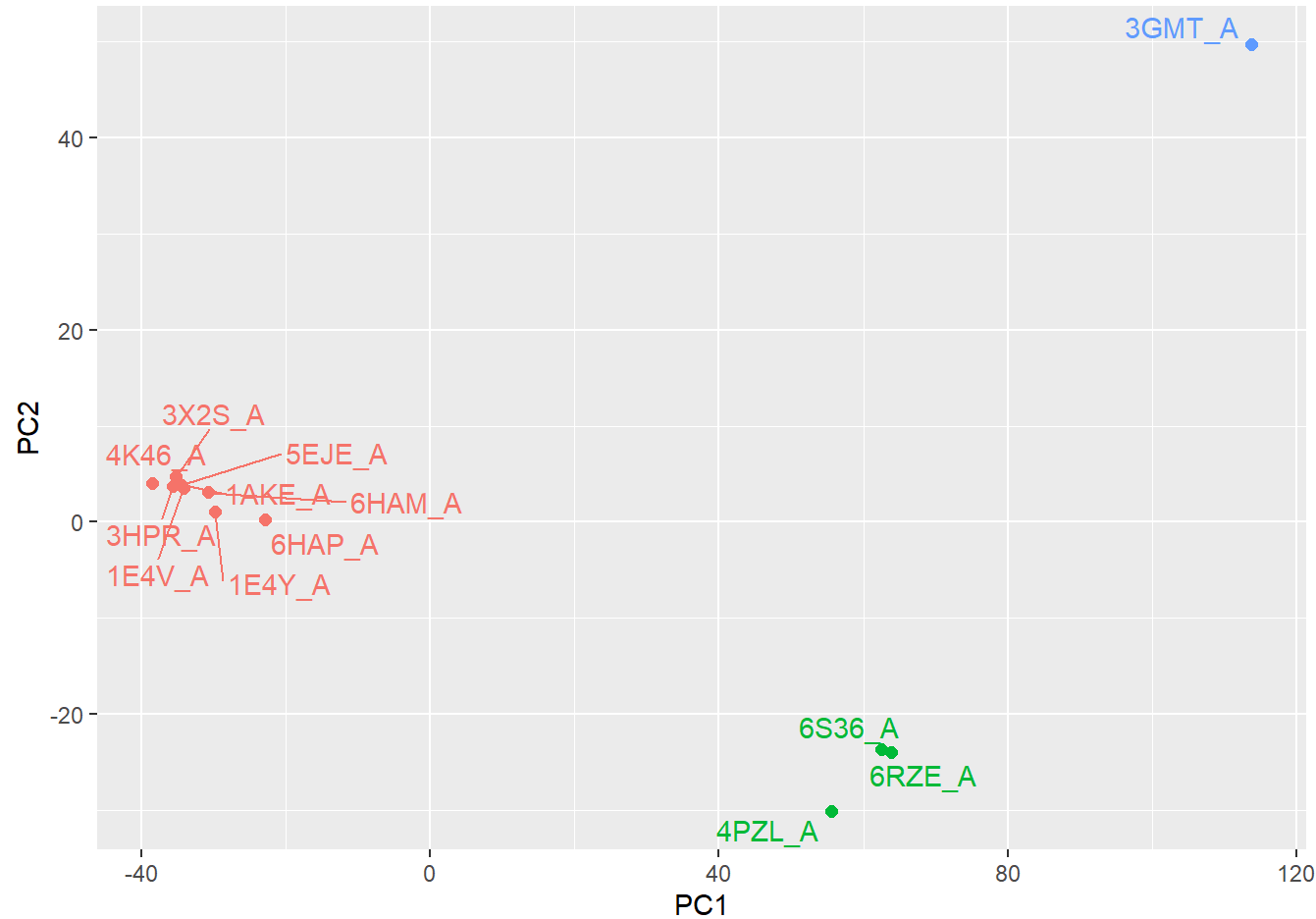


PCA Visualization

```
pc1 <- mktrj(pc.xray, pc = 1, file = "pc_1.pdb")
```

Plotting PCA Results with GGPLOT2

```
df <- data.frame(  
  PC1 = pc.xray$z[, 1],  
  PC2 = pc.xray$z[, 2],  
  col = as.factor(grps.rd),  
  ids = ids  
)  
  
p <- ggplot(df) +  
  aes(PC1, PC2, col = col, label = ids) +  
  geom_point(size = 2) +  
  geom_text_repel(max.overlaps = 20) +  
  theme(legend.position = "none")  
  
p
```



Normal Mode Analysis

```
modes <- nma(pdb)
```

Details of Scheduled Calculation:

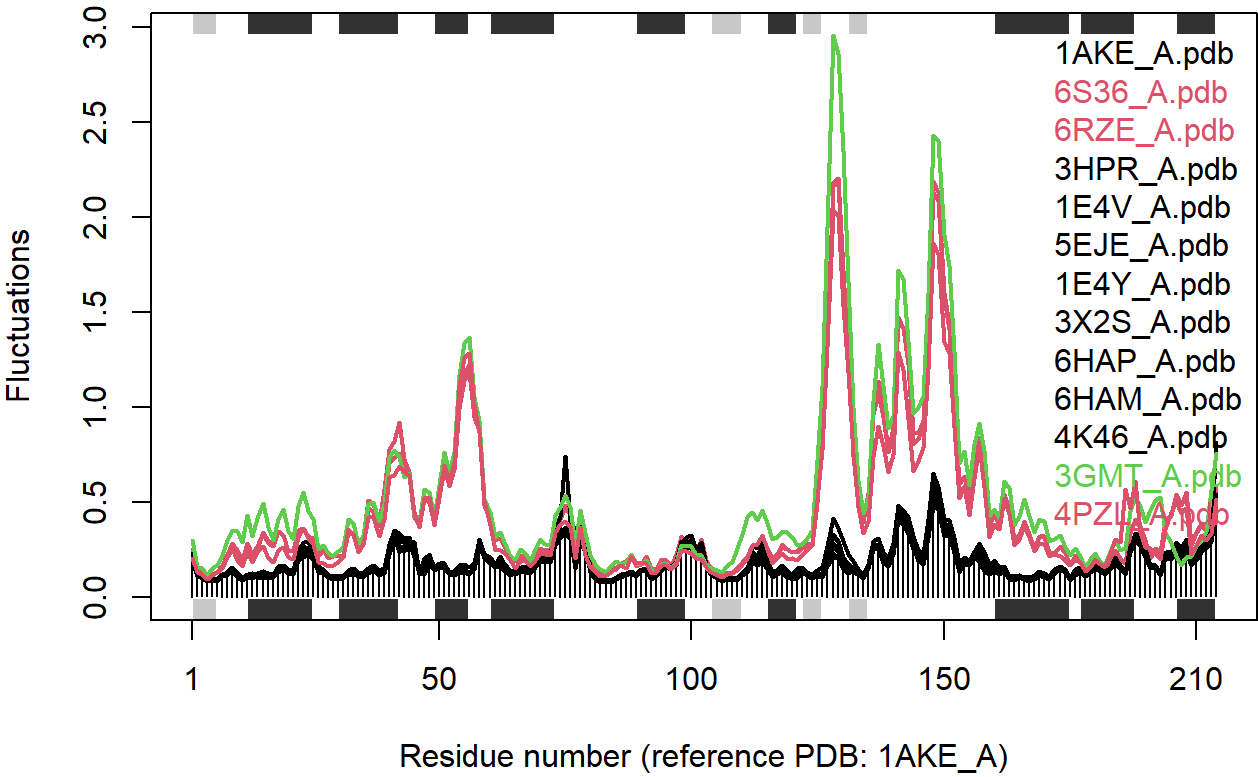
- ... 13 input structures
- ... storing 606 eigenvectors for each structure
- ... dimension of x\$U.subspace: (612x606x13)
- ... coordinate superposition prior to NM calculation
- ... aligned eigenvectors (gap containing positions removed)
- ... estimated memory usage of final 'eNMA' object: 36.9 Mb

		0%
=====		8%
=====		15%
=====		23%



```
plot(modes, pdb, col = grps.rd)
```

Extracting SSE from pdb\$sse attribute



Q14: What do you note about this plot?

The black and colored lines are not identical. The colored lines show larger fluctuations in certain regions, especially around the flexible nucleotide-binding regions of adenylate kinase. These regions likely differ the most because they are involved in conformational changes between open and closed states during ligand binding and enzyme function.